AC-75 Rotary Screw Air Compressor: Service Manual

Document	Revision	Effective	Owner
NG-MAN-AC75	C	2025-03-01	Northgate Equipment Co.

1. Overview

The Northgate AC-75 is a 75 HP oil-flooded rotary screw air compressor. It supplies plant air for conveyor diverters, palletizers and dock equipment. The controller reports discharge temperature, line pressure, power draw and airend vibration.

2. Specifications

Item	Value
Motor	75 HP, 460V / 3 ph, TEFC
Free air delivery	322 CFM at 125 psig
Pressure setpoint	125 psig, load/unload band 115 to 135 psig
Lubricant	Northgate SynLube 8000, 9 gal
Bearings	Airend: tapered roller; motor: 6316 drive end, 6313 non-drive end

3. Normal Operating Ranges

Measurement	Normal range	Alarm limit
Discharge temperature	165F to 195F	Above 220F (T-05), shutdown at 230F
Line pressure	115 to 135 psig	Below 100 psig
Power draw	48 to 60 kW loaded	Above 68 kW
Airend vibration	Below 4.5 mm/s RMS	4.5 mm/s alert (V-02), 7.1 mm/s alarm (V-03)

Vibration limits follow ISO 10816-3 for medium machines on rigid foundations. A steady rise over days is more important than a single high reading. Trend vibration weekly.

Fault Code Reference

Code	Name	Trigger condition	Severity
V-02	Vibration alert	Airend vibration above 4.5 mm/s RMS for 1 hour.	Medium
V-03	Vibration alarm	Airend vibration above 7.1 mm/s RMS.	High
T-05	High discharge temperature	Discharge temperature above 220F. Controller shuts down at 230F.	High
P-07	Low oil pressure	Oil injection pressure below 20 psig while loaded.	High

Troubleshooting by Fault Code

V-02: Vibration alert

Trigger: Airend vibration above 4.5 mm/s RMS for 1 hour.

Likely causes, most common first:

- Early bearing wear.
- Coupling misalignment after maintenance.
- Loose mounting bolts.

Corrective actions:

1. Check mounting bolt torque (85 ft-lb).
2. Check coupling alignment with a laser tool. Limit is 0.002 in offset.
3. Take an oil sample for wear-metal analysis.
4. Increase vibration readings to daily until the trend is flat.

V-03: Vibration alarm

Trigger: Airend vibration above 7.1 mm/s RMS.

Likely causes, most common first:

- Advanced bearing wear (most common when vibration rose steadily over 1 to 2 weeks).
- Severe misalignment.
- Rotor contact inside the airend.

Corrective actions:

1. Plan a controlled shutdown within 24 hours. Switch plant load to the backup compressor first.
2. Apply LOTO, then bleed the receiver tank to 0 psig before any work.
3. Inspect motor bearings for noise and heat. Replace the 6316 drive-end bearing if rough.
4. Send oil for analysis. Iron above 50 ppm confirms bearing or rotor wear.
5. After repair, run 4 hours and confirm vibration below 4.5 mm/s.

Running above 7.1 mm/s risks airend failure. A replacement airend has a 6 to 8 week lead time.

T-05: High discharge temperature

Trigger: Discharge temperature above 220F. Controller shuts down at 230F.

Likely causes, most common first:

- Low oil level.
- Clogged oil cooler.
- Failed thermal valve.
- Room temperature above 104F.

Corrective actions:

1. Check oil level on the sight glass.
2. Clean the oil cooler fins.
3. Test the thermal valve.
4. Improve room ventilation.

P-07: Low oil pressure

Trigger: Oil injection pressure below 20 psig while loaded.

Likely causes, most common first:

- Clogged oil filter.
- Low oil level.

Corrective actions:

1. Replace the oil filter.
2. Top up oil to the middle of the sight glass.

Preventive Maintenance

Task	Interval
Vibration route reading	Monthly; weekly if above 3.5 mm/s
Oil sample analysis	Every 2,000 run hours
Replace oil filter and separator	Every 4,000 run hours
Grease motor bearings	Every 2,000 run hours, 30 g per bearing

WARNING: Apply lockout/tagout per SOP-OPS-001 before opening any electrical enclosure or touching moving parts. Only authorized technicians may perform these steps. Stored air pressure is a hazardous energy source. Bleed the receiver to 0 psig.